

BMEG3105 Data Analytics for Personalized Genomics and Precision Medicine

Scribing: Lec-4 (Assembly and Mapping)

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Note: This concise scribing follows the layout and style of the supplied BMEG3105 examples. Figures and tables are adapted from those examples and the lecture materials.

Outline of Lecture

- More about dynamic programming
- Gene expression matrix
- Sequence assembly and sequence mapping

0 Recap from Last Lecture

Dynamic programming (DP) finds an optimal sequence alignment by breaking the problem into smaller sub-problems. The DP table stores both scores and the path used to construct the alignment. This problem-solving process is more important than memorizing one implementation.

1 More about Dynamic Programming

1.1 Procedure

1. Prepare a scoring matrix.
2. Fill the DP table and preserve arrows.
3. For global alignment, the best score is in the last cell.
4. Trace back the arrows to obtain an optimal alignment.

	A	C	G	T
A	2	-7	-5	-7
C	-7	2	-7	-5
G	-5	-7	2	-7
T	-7	-5	-7	2

Gap penalty = -10

		A	C	C	G
	0	-10	-20	-30	-40
A	-10	2	-8	-18	-28
C	-20	-8	4	-6	-16
G	-30	-18	-6	-3	-4

For ACCG and ACG, $F(4,3) = -4$. Two optimal alignments are ACCG/A _CG and ACCG/AC _G.

		A	C	C	G
	0	-10	-20	-30	-40
A	-10	2	-8	-18	-28
C	-20	-8	4	-6	-16
G	-30	-18	-6	-3	-4

Dynamic-programming table and traceback example from the reference scribing.

1.2 Global, Local, and Scoring Concepts

Global alignment aligns sequences end to end. Local alignment searches for the best matching subsequences, which is useful for finding motifs or domains.

Biological meaning. Mismatches may represent mutations, while gaps may represent insertions, deletions, or gene duplications. A scoring matrix must match the purpose of the comparison. BLOSUM matrices, for example, score protein alignments using local evolutionary information.

Concept	Example interpretation
Match	Similarity between bases or amino acids
Mismatch	Possible mutation
Gap	Insertion, deletion, or duplication
BLOSUM	Protein substitution scores for divergent sequences

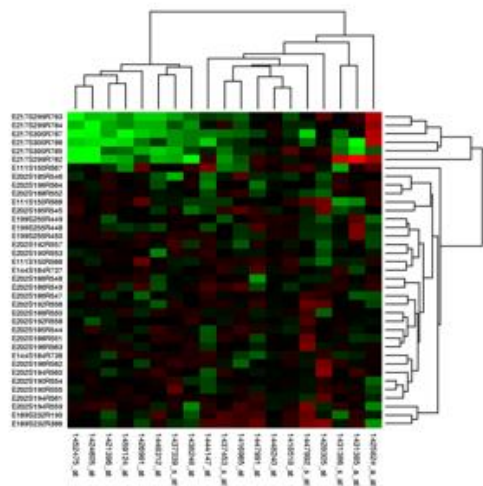
Efficiency. Enumerating all alignments requires $C(2n,n)$ possibilities, about 7×10^{88} when $n=300$. DP reduces the work to about n^2 table entries, or 90,000 for $n=300$.

2 Why and How to Get a Gene Expression Matrix

The central dogma is DNA → RNA → protein. Phenotype depends on genotype and environment. Since human genomes are highly similar and only about 1% encodes protein, differences in gene expression can provide information about phenotype differences.

2.1 Gene Expression Matrix

Rows represent genes, columns represent samples or cells, and color intensity represents expression level. Clustering groups genes or samples with similar expression patterns.

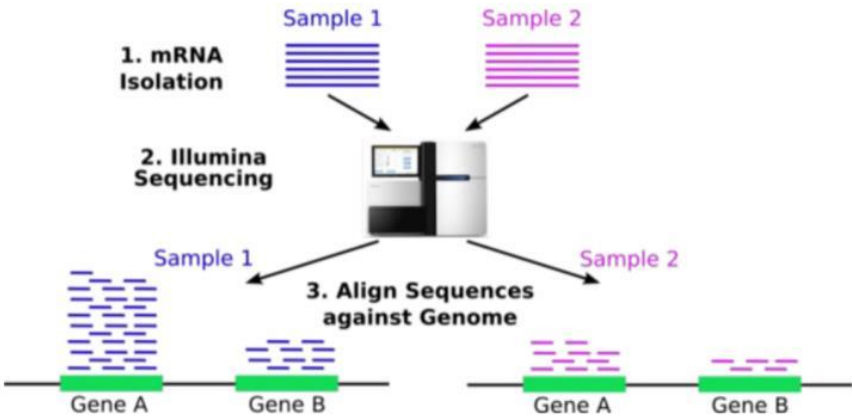


Gene-expression heatmap with hierarchical clustering from the reference scribing.

Matrix element	Meaning
Rows	Genes
Columns	Samples or cells
Color	Expression level
Clustering	Similarity or distance

2.2 Conversion Pipeline

- 5. Isolate mRNA from samples.
 - 6. Sequence short reads using Illumina sequencing.
 - 7. Map reads to a reference genome.
 - 8. Count reads assigned to each gene.
 - 9. Normalize for gene length and organize counts into a matrix.
- Goal:** map short reads to the genome and count them to estimate gene expression.



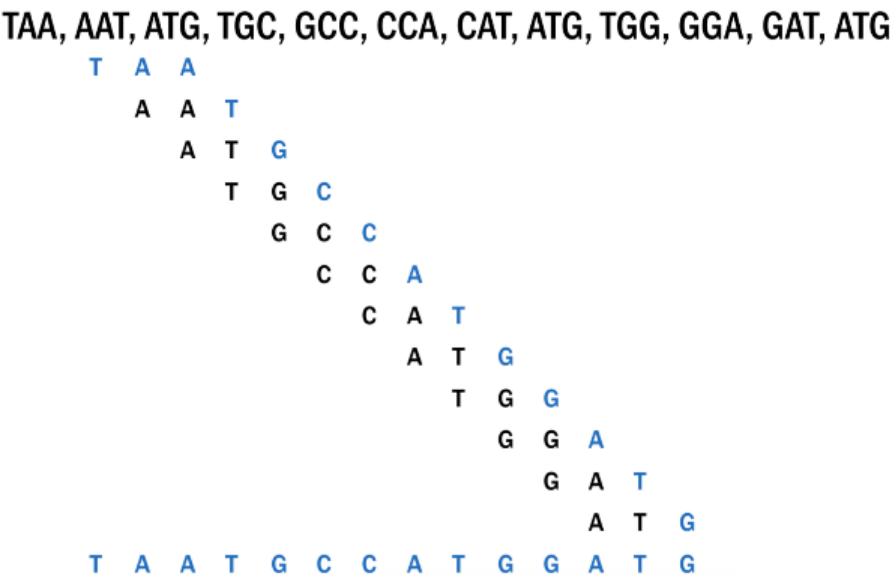
Overview of the mRNA sequencing and gene-expression workflow.

3 Sequence Assembly and Sequence Mapping

3.1 Genome Assembly

Genome assembly uses overlaps among short reads to reconstruct a longer sequence. The human genome is about 3×10^9 bp, while an Illumina read is about 200 bp.

Example reads: TAA, AAT, ATG, TGC, GCC, CCA, CAT, ATG, TGG, GGA, GAT, ATG. Overlapping them gives TAATGCCATGGATG.



Overlap-based assembly example from the reference scribing.

Main difficulties: mutations can remove or create overlaps; conflicting reads may require a choice; repeats obscure true length; repeated units obscure copy number; and large datasets require fast algorithms.

3.2 Sequence Mapping

Mapping slides a read along the reference genome and calculates the difference at each position. The smallest difference indicates the best match. For genome TAATGCCATGGATG and read CCA, the minimum difference is 0 at position 6.

Position	1	2	3	4	5	6	7	8	9	10	11	12
Difference	2	3	3	3	2	0	2	3	3	2	3	3



Read-mapping difference example from the reference scribing.

If CCA is mapped three times, its **gene-expression count** is 3. Questions include how to improve speed, map reads in one pass, and handle mutations or sequencing errors.

3.3 BLAST

BLAST (Basic Local Alignment Search Tool) finds short local matches, extends them, and combines high-scoring regions. For query ACGTTGCT, it may first search for ACG and similar words such as CCG or GCG, then extend the match until the score falls below a threshold.

Potential Project-1

- Find a reference genome and sequencing reads.
- Map reads to the reference genome.
- Count reads for each gene.
- Explain the software, data, and every pipeline step in the report.

Summary

- DP efficiently finds optimal alignments and records the construction path.
- Scoring matrices determine the biological meaning of matches, mismatches, and gaps.
- Gene expression matrices summarize read counts across genes and samples.
- Assembly connects overlapping reads; mapping places reads on a reference genome.
- BLAST searches for and extends local sequence matches.

Useful Links

- Biopython: <https://biopython.org>
- BLAST: <https://blast.ncbi.nlm.nih.gov/Blast.cgi>
- EMBOSS Needle: https://www.ebi.ac.uk/Tools/psa/emboss_needle/
- VGP assembly pipeline: <https://github.com/VGP/vgp-assembly/tree/master/pipeline>